

Test Planner and Tracker - Team 28

Team Fleetix

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Test ID	Related Use case	Pre-condition(s)	Test Description	Expected Outcome	R1 Outcome	Comments (if test case failed)	R2 Outcome	Comments (if test case failed)
TC-001	UC001 - Geocoding	System is operational. Address strings are available for processing and are recognizable by the OSM database.	1. Enter multiple address strings into the system 2. System processes these addresses through the Geocoding API 3. Verify coordinates are returned	Returns valid geographic coordinates (latitude, longitude), that are accurate (within ~500 meters of the real-world location).		NA		NA
TC-002	UC004 - Routing (Many-to-One)	System is operational. Source and destination coordinates are valid.	1. Enter a valid source address 2. Enter a valid destination address 3. Request route generation 4. Verify route details	A route is generated along with details like total distance (in km), estimated duration (based on historic data). The route must follow real-world roads and minimise time.		NA		NA
TC-003	UC004 - Routing (Many-to-One)	System is operational. Source, destination, and multiple pickup coordinates are valid.	1. Enter a valid source address 2. Enter a valid destination address 3. Enter multiple pickup locations 4. Request route generation 5. Verify route details and ordering	Along with the expectations from TC-002, this use case should return the ordered lists of the waypoints that minimises the overall time taken.		NA		NA
TC-004	UC008 - Routing (Many-to-Many)	System is operational. Multiple source and destination coordinates are valid. Multiple agents (vehicles) are available.	1. Define multiple pickup locations 2. Define multiple destination locations 3. Specify number of available vehicles 4. Request route optimization 5. Verify route assignments and timing	Along with the expectations from TC-002, this use case should return the ordered lists of the waypoints that minimises the overall time taken for each agent. The distribution between the agents should be such that the time taken by the maximum-timed agent is minimised.		NA		NA
TC-005	UC005 - Visualise Route	System is operational. Routes have been generated and are available for visualization.	1. Verify visual elements 2. Test map interaction features.	Route is displayed on a map with markers to clearly signify source, destination, and pickups. Visuals must include colour-coded (in case of many-to-many) path overlay Color coding per agent. Also provide map interaction like zoom, pan, move around, etc		NA		NA
TC-006	UC009 - User Interface	System is operational.	Intuitive application design with optimised user interface for improved and efficient user experience	Accessed the system through web interface Navigated through main sections Verified responsiveness and functionality		NA		NA
TC-007	UC009 - User Interface	System is operational. Address input component is available.	1. Begin typing an address in the address input field 2. Observe system suggestions 3. Time the response 4. Select a suggestion and verify autofill	System returns a list of suggested complete address options ordered by relevance. Dynamically updates as the user types and autofills when user selects. Fast response time of <0.5s		NA		NA

TC-008	UC009 - User Interface	System is operational. User management module is implemented.	1. Create accounts for each user role (admin, driver, employee) 2. Attempt to access role-specific features with each account 3. Verify appropriate access controls	Each user role has access to their appropriate interface and features. Admins can manage users, drivers can view routes, and employees can set preferences.		Not implemented in R1		NA
TC-009	UC003 - Employee Pick-up & Drop-off Location Preference	System is operational. Employee accounts are created.	1. Log in as an employee 2. Navigate to preferences section 3. Set preferred pickup location 4. Set preferred drop-off location. 5. Save preferences 6. Generate a route that includes this employee	Employee can successfully set and update pickup/drop-off preferences. These preferences are saved to their account and used when generating routes.		Not implemented in R1		NA
TC-010	UC006 - Adding Constraints	System is operational. Admin account is created.	1. Log in as admin 2. Navigate to constraint management 3. Add various constraints (vehicle capacity, time windows, maximum route duration) 4. Generate routes with these constraints 5. Verify constraints are applied	Admin can successfully add and manage various constraints. These constraints are properly applied when generating routes.		Not implemented in R1		NA
TC-011	UC007 - Ride History and Reports	System is operational. Previous rides data is available.	1. Log in as admin or employee 2. Navigate to ride history section 3. View past rides 4. Filter rides by date, location 5. Generate reports	User can view past rides with details including route, timing. Filtering options work correctly and reports can be generated.		Not implemented in R1		NA
TC-012	UC009 - User Interface	System is operational. Admin can add, edit and remove an employee or a driver details	1. Log in as admin 2. Navigate to user management 3. Perform the CRUD operation 5. Verify all changes are applied throughout	Admin can successfully add and manage all the non-admin users in the database.		Not implemented in R1		NA
TC-013	UC009 - User Interface and UC004 - Routing (Many-to-One)	System is operational. Route has been generated.	1. Generate optimized route. 2. Save Route to database 3. Verify driver assignment 4. Confirm all route details are preserved	Route is saved with all details intact. Driver is assigned if available. System provides feedback on successful save and driver assignment status.		Not implemented in R1		NA
TC-014	UC004 - Routing (Many-to-One)	System is operational. Multiple drivers are registered in the system with varying availability states.	1. Query driver availability for a specific time window. 2. Check availability for drivers with overlapping routes 3. Verify accuracy of availability status	System correctly identifies available and unavailable drivers for the specified time period, considering their current route assignments and rest periods		Not implemented in R1		NA
TC-015	UC004/UC008 - Routing	System is operational. Multiple drivers are available. Multiple routes need assignments.	1. Generate multiple routes requiring driver assignment 2. Observe driver assignment distribution 3. Check driver counts assigned to each	Workload is fairly distributed among drivers, with priority given to drivers with fewer assignments (drivesCount). No driver is overloaded while others remain underutilized.		Not implemented in R1		NA
TC-016	UC006 - Adding Constraints	System is operational. Vehicle capacity constraint is set.	1. Set vehicle capacity constraint 2. Generate route with more pickups than vehicle capacity 3. Observe route generation behavior	System properly divides pickups among multiple vehicles based on capacity. No vehicle is assigned more pickups than its capacity.		Not implemented in R1		NA
TC-017	UC001 - Geocoding	System is operational.	1. Enter multiple invalid or ambiguous addresses 2. Observe system response and error handling 3. Test system recovery options	System provides clear error messages for invalid addresses.		Not implemented in R1		NA

